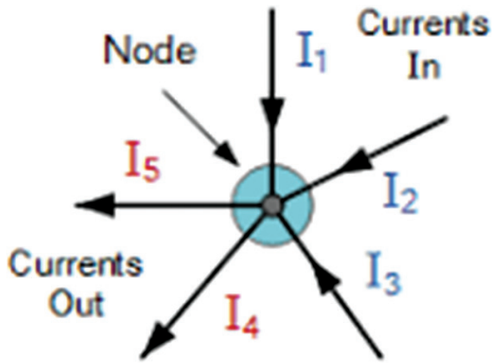


Fig 2.8(i) *Kirchhoff's Current Law*

the currents at any junction of a network is zero.

Fig 2.8(i) represents Kirchhoff's Current Law. In this, I_1, I_2 and I_3 represent the current flowing towards the junction point I_4 and I_5 represent the current flowing away from the junction point.

$$I_1 + I_2 + I_3 = I_4 + I_5$$

$$I_1 + I_2 + I_3 - I_4 - I_5 = 0$$

ii) Kirchhoff's Voltage Law (KVL)

In any closed electric circuit, the algebraic sum of the potential drop is equal to the supply voltage.

Fig 2.8(ii) represents, loads R_1, R_2, and R_3 connected in series. Potential drop across R_1 is IR_1, potential drop across R_2 is IR_2 and potential drop across R_3 is IR_3.

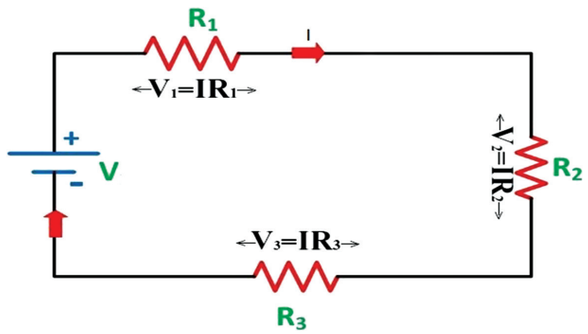


Fig 2.8(ii) *Kirchoff's Voltage Law*

$$V_1 = IR_1$$
$$V_2 = IR_2$$
$$V_3 = IR_3$$
$$V = V_1 + V_2 + V_3$$
$$IR = IR_1 + IR_2 + IR_3$$



Gustav Robert Kirchhoff

Born: 1824

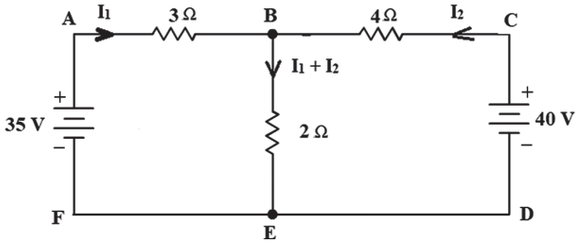
Died: 1887

Gustav Robert Kirchhoff (1824-1887) was a German physicist who contributed to the fundamental understanding of electrical circuits. Kirchhoff formulated his circuit laws, which are now ubiquitous in electrical engineering.

Problems:

Example: 1

Calculate the current in a 2Ω resistor using Kirchoff's law in the following circuit.



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